

APPENDIX A THE DISTRIBUTION OF NEWS

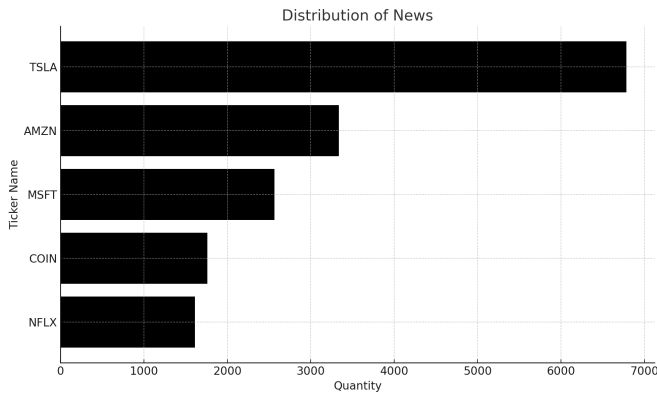


Fig. 11: The distribution of news collected from the Alpaca News API for the five stocks used in the experiments.

APPENDIX B DETAILED INTRODUCTION ABOUT THE COMPARATIVE MODELS

DRL trading agents:

As the FINMEM is practiced and examined on the basis of single stock trading and discrete trading actions, we choose three advanced DRL algorithms fitting into the same scenarios according to the previous and shown expressive performance in the work of Liu et al. [31, 29]. The DRL training agents only take numeric features as inputs.

- **Proximal Policy Optimization (PPO):** PPO [50] is employed in stock trading due to its stability and efficiency. One salient advantage of PPO is that it maintains a balance between exploration and exploitation by bounding the policy update, preventing drastic policy changes.
- **Deep Q-Network (DQN):** DQN [36] is an adaptation of Q-learning, that can be used to optimize investment strategies. Unlike traditional Q-learning that relies on a tabular approach for storing Q-values, DQN generalizes Q-value estimation across states using deep learning, making it more scalable for complex trading environments.
- **Advantage Actor-Critic (A2C):** A2C [35] is applied to optimize trading actions in the financial environment. It operates by simultaneously updating both the policy (actor) and the value (critic) functions, providing a balance between exploration and exploitation.

LLM trading agents:

- **General-purpose Generative Agents – GA:** The generative AI agent by Park et al. [44], originally intended to simulate realistic human behavior and make everyday decisions, has been adapted here for specific stock trading tasks. This agent’s architecture includes a memory module that employs recency, relevance, and importance metrics to extract pivotal memory events for informed decision-making. However, it does not provide a layered memory module to effectively differentiate the time

sensitivities unique to various types of financial data. Additionally, although it features a profile module to define agent attributes like professional background, the model does not include a mechanism for self-adaptive risk preference. In our experiments, we modified the original prompt template created by Park et al., which was intended for general daily tasks, to suit financial investment tasks. The textual elements of this revised template closely align with those of FINMEM, with the exception of two components that are absent in this version of general-purpose Generative Agents.

- **LLM trading agents – FINGPT:** A novel open-source LLM framework specialized for converting incoming textual and numeric information into informed financial decision-making, introduced by Yang et al. [62]. It claims superiority over the traditional buy-and-hold strategy.

APPENDIX C SENSITIVITY ANALYSIS: IMPACT OF MEMORY DECAY SPEED ON FINMEM’S TRADING PERFORMANCE

In our fourth ablation study, we evaluated the effectiveness of FINMEM’s memory ranking and retrieval mechanisms, as detailed in Section III-C. We specifically examined how varying memory decay rates in the *shallow processing layer of the long-term memory data warehouse* affected trading performance. This layer primarily contained investment insights extracted from market news. We chose this layer as the focus due to its higher volume and frequency of memory events, making the impact of changes more readily observable. Events in the intermediate and deep processing layers, which primarily comprised insights from quarterly and annual financial reports, could significantly influence FINMEM’s decisions. However, their effects were often less easily observable due to their lower frequency and quantity, as well as their integration with events escalated from the shallow layer.

We adjusted the decay parameters for both Recency and Importance scores, as defined in Equations 2 and 5 respectively. The values presented in Table V were calibrated to ensure that memory events in the shallow layer would reach the lower threshold for memory cleanup after specific periods, as measured in days. For either the Recency or Importance score, the cleanup threshold was set at 0.05 (normalized to a 0-1 scale). We analyzed the Cumulative Returns and Sharpe Ratios under different memory decay speeds, as illustrated in Figure 12. Our findings revealed two key points: 1) Aligning the memory decay speed with the actual timeliness of specific types of memory events enhances FINMEM’s decision quality. This appropriate calibration ensures that relevant information is retained and utilized effectively. 2) The original setting for the shallow memories decay rate, which retains market news in FINMEM’s long-term memory for 14 days, proved optimal for performance. This timeframe allows FINMEM to conduct further operations on the retained information such as market news in a reasonably effective period, leading to the best trading outcomes. These results underscore the nuanced design of FINMEM’s memory module in processing information with varying degrees of timeliness. And they highlight memory

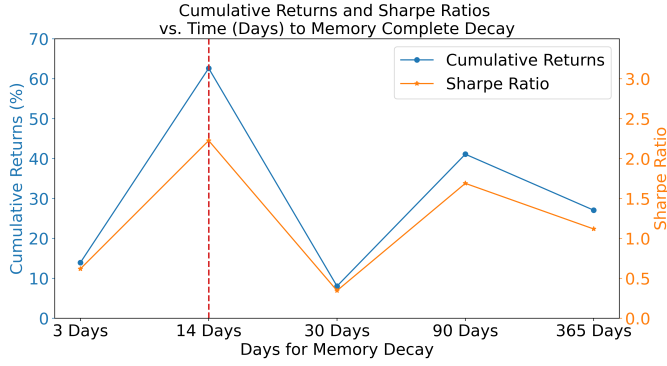


Fig. 12: Performance comparison of FINMEM across various shallow layer decay speeds in its long-term memory component, measured by CRs and SRs. The optimal performance was achieved using the default decay setting, which applies decay over a 14-day period, as implemented in the main experiments shown in Section V-B.

mechanisms are tuned with careful consideration about the nature of the financial trading environment.

Time Length (Days) of Complete Decay	Recency Decay ($Q_{shallow}$)	Importance Decay ($\alpha_{shallow}$)
3	1.0	0.5000
14	3.0	0.9200
30	10.02	0.9330
90	30.04	0.9771
365	120.0	0.9973

TABLE V: Parameter options for recency and importance scores ($S_{Recency_{shallow}}^E$ and $S_{Importance_{shallow}}^E$) to decay in the required time lengths. The selection of these parameters directly influences the values of the information retrieval score $\gamma_{shallow}^E$ and the memory event rankings in FINMEM’s cognitive system.

APPENDIX D

THE RISK INCLINATION REQUIRED FOR FINMEM TO ACHIEVE OPTIMAL TRADING PERFORMANCE VARIES BY INDIVIDUAL STOCK.

	Risk-Averse	Self-Adaptive	Risk-Seeking
TSLA		✓	
NFLX		✓	
AMZN		✓	
MSFT			✓
COIN		✓	

Fig. 13: The risk inclination required for FINMEM to achieve optimal trading performance varies by individual stock. FINMEM attained its best overall performance using a self-adaptive risk setting, whereas it achieved peak performance for MSFT under a risk-seeking setting.